3.0.12

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Question

A loaded dice has following probability distribution of occurrences

Dice value	1	2	3	4	5	6
Probability	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$

If three identical dice as the above are thrown, the probability of occurrence of values $1, 5$ and 6 on the three dice is (EE 2007)

(a) same as that of occurrence of $3, 4, 5$

(b) same as that of occurrence of $1, 2, 5$

(c) $\frac{1}{128}$

(d) $\frac{5}{8}$

solution

The Probability of occurance of 1, 5, 6:
let,
A = Event that 1 appeared when the die with the given probability distribution is thrown.
B = Event that 5 appeared when the die with the given probability distribution is thrown.
C = Event that 6 appeared when the die with the given probability distribution is thrown.

$$P(A) = 1/4 \tag{1}$$
$$P(B) = 1/8 \tag{2}$$
$$P(C) = 1/4 \tag{3}$$

Occurence of 1-6 on each of the dice is independent of the output of the other.
$\therefore$A, B, C are independent of each of other.

$$P(A \cap B \cap C) = P(A)P(B)P(C) \tag{4}$$
$$= (1/4)(1/8)(1/8) \tag{5}$$
$$= 1/128 \tag{6}$$

$\therefore$ The Probability of occurance of 1, 5, 6=1/128

Option A:

Occurance of 3,4,5:

A = Event that 3 appeared when the die with the given probability distribution is thrown.

B = Event that 4 appeared when the die with the given probability distribution is thrown.

C = Event that 5 appeared when the die with the given probability distribution is thrown.

$$P(A \cap B \cap C) = P(A)P(B)P(C) \tag{7}$$
$$= (1/8)(1/8)(1/8) \tag{8}$$
$$= 1/512 \tag{9}$$

Option B:

Occurance of 1,2,5:

A = Event that 1 appeared when the die with the given probability distribution is thrown.

B = Event that 2 appeared when the die with the given probability distribution is thrown.

C = Event that 5 appeared when the die with the given probability distribution is thrown.

$$P(A \cap B \cap C) = P(A)P(B)P(C) \tag{10}$$
$$= (1/4)(1/8)(1/8) \tag{11}$$
$$= 1/256 \tag{12}$$

Option (C) is correct.

C Code

```c
#include <stdio.h>

int main() {
    double prob[7]; // Indices 1-6 for ease of use
    int v1, v2, v3;
    double result;

    // 1. Input the probabilities for sides 1 to 6
    printf("Enter the probabilities (as decimals, e.g., 0.125 for
         1/8):\n");
    for (int i = 1; i <= 6; i++) {
        printf("Side %d: ", i);
        scanf("%lf", &prob[i]);
    }
```

C Code

```c
// 2. Input the three values you want to check
printf("\nEnter the 3 values to calculate (e.g., 1 5 6): ");
if (scanf("%d %d %d", &v1, &v2, &v3) != 3) {
    printf("Error: Please enter three integers.\n");
    return 1;
}
```

C Code

```
    // 3. Calculation for the specific sequence (Intersection of
        independent events)
    // Formula: P(v1) * P(v2) * P(v3)
    result = prob[v1] * prob[v2] * prob[v3];

    printf("Probability of sequence (%d, %d, %d): %.10f\n", v1,
        v2, v3, result);
    return 0;
}
```

Python code

```
import numpy as np
import matplotlib.pyplot as plt
# 1. Define the PMF
faces = np.array([1, 2, 3, 4, 5, 6])
probabilities = np.array([1/4, 1/8, 1/8, 1/8, 1/8, 1/4])
# 2. Plot the PMF
plt.figure(figsize=(8, 5))
plt.stem(faces, probabilities, basefmt=" ")
plt.title("Probability Mass Function (PMF) of Loaded Die")
plt.xlabel("Die Value")
plt.ylabel("Probability")
plt.xticks(faces)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

Python code

```
num_simulations = 1_000_000
# Roll 3 dice simultaneously for each simulation
rolls = np.random.choice(faces, size=(num_simulations, 3), p=
    probabilities)

# Check for specific sequence (1, 5, 6)
successes = np.all(rolls == [1, 5, 6], axis=1)
numerical_prob = np.mean(successes)

print(f"Theoretical Probability (1, 5, 6): {1/128:.6f}")
print(f"Numerical Probability (1, 5, 6): {numerical_prob:.6f}")
```

Plot

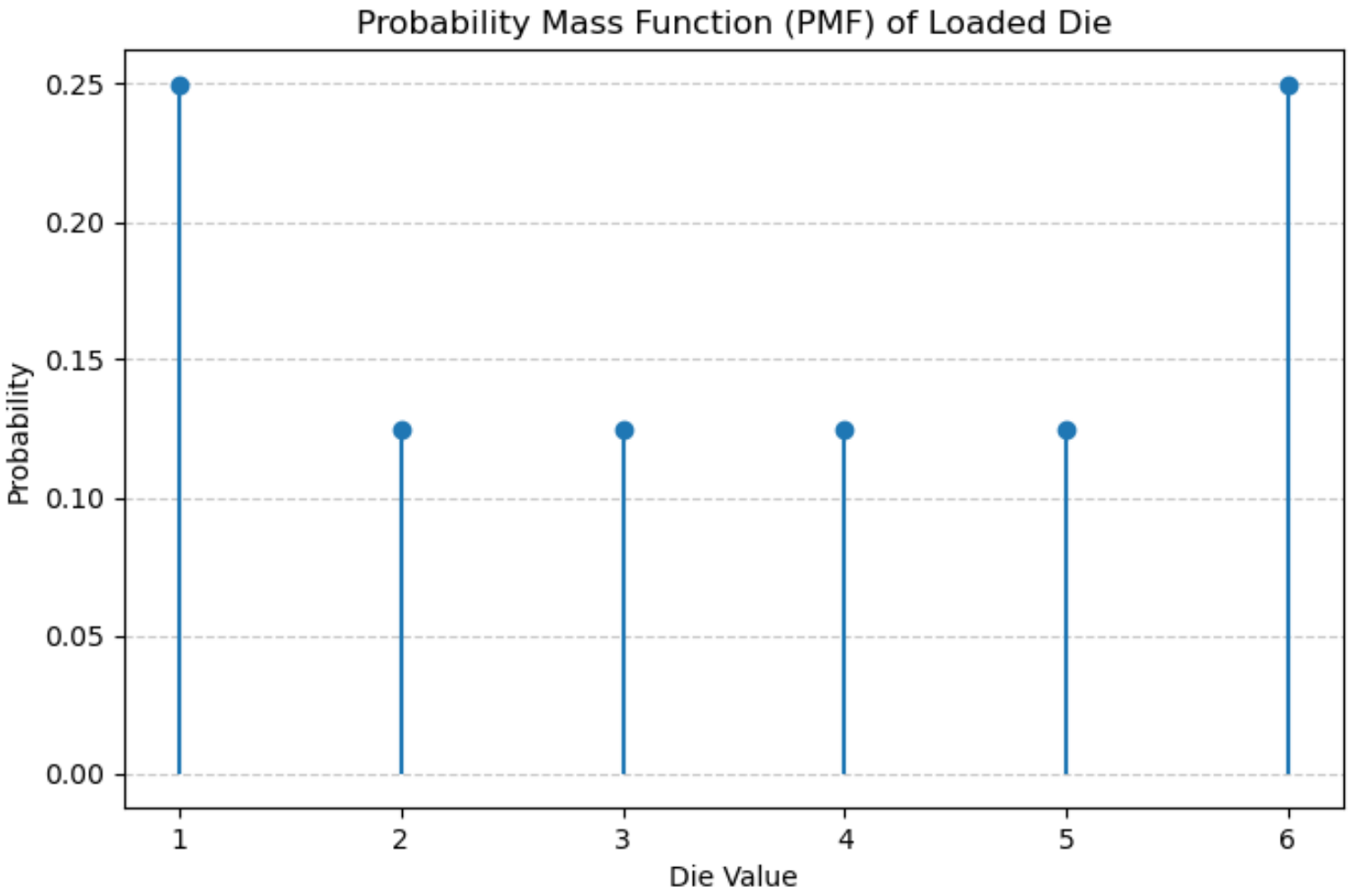

Figure: PMF

Numerically Calculated Probability

The desired Probability is computed numerically by generating the distribution by rolling the dice 1000000 times.

```
madhuri_beeram@Madhuri:~$ python3 pmf.py
Theoretical Probability (1, 5, 6): 0.007812
Numerical Probability (1, 5, 6):   0.007787
```

Figure: Output

Numerically found value closely matches with the calculated Probability.